Analog matrix LC1

Circuit simulation review | 2 October 2026

A separate 12-bit, lower-cost alternative to Revision D. Both designs are unbuilt prototypes. This report covers the circuit model; PCB design-rule evidence is supplied separately.

Target	LC1 implementation
Signal path	4 inputs x 4 outputs; signed coefficients; DC through 1 MHz
Amplitude / load	Inputs +/-1 V peak; outputs +/-4 V peak into external 50 ohm. Internal driver swing approaches +/-8 V.
Coefficient control	16 DAC7811 multiplying DACs, eight OPA2810 duals, ten TMUX6136 dual switches; original ADA4899 TIAs and THS3091 output drivers retained.
Offset correction	MCP4728 + OPA4197; 100 us analog filters; 2 ms muted settling interval.

What the results establish

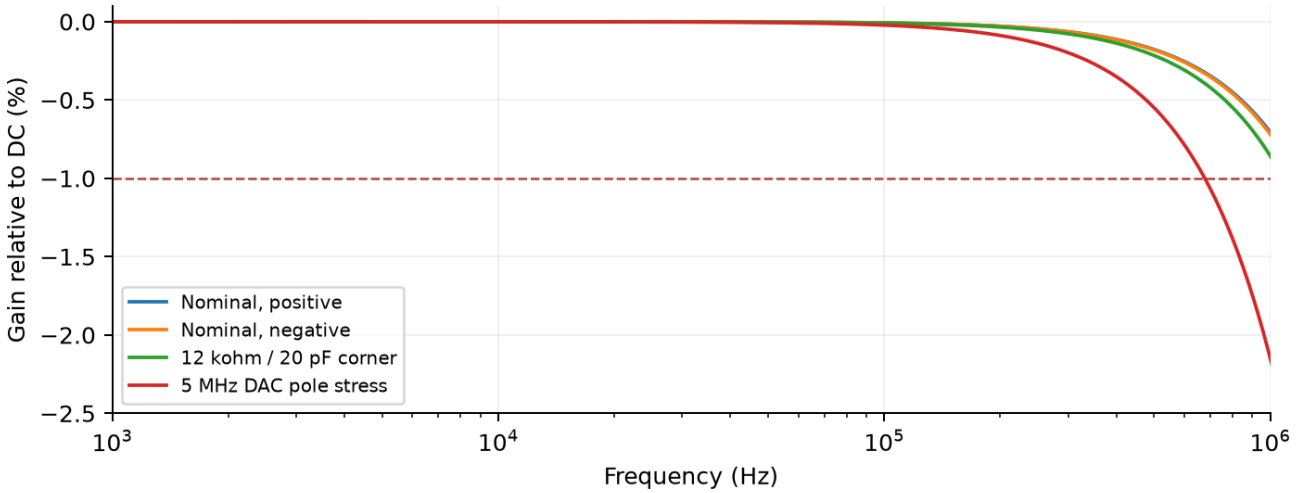
35 recorded benches completed: 13 row transfer/large-signal cases and 22 feedback-loop cases. Netlist SHA-256 values match the recorded results. Nominal magnitude droop at 1 MHz is 0.70%; the modeled resistor/capacitance corner reaches 0.86%. Minimum modeled phase margin is 57.5 degrees. Firmware protocol and failure-muting tests pass on a host simulator.

What remains to be measured

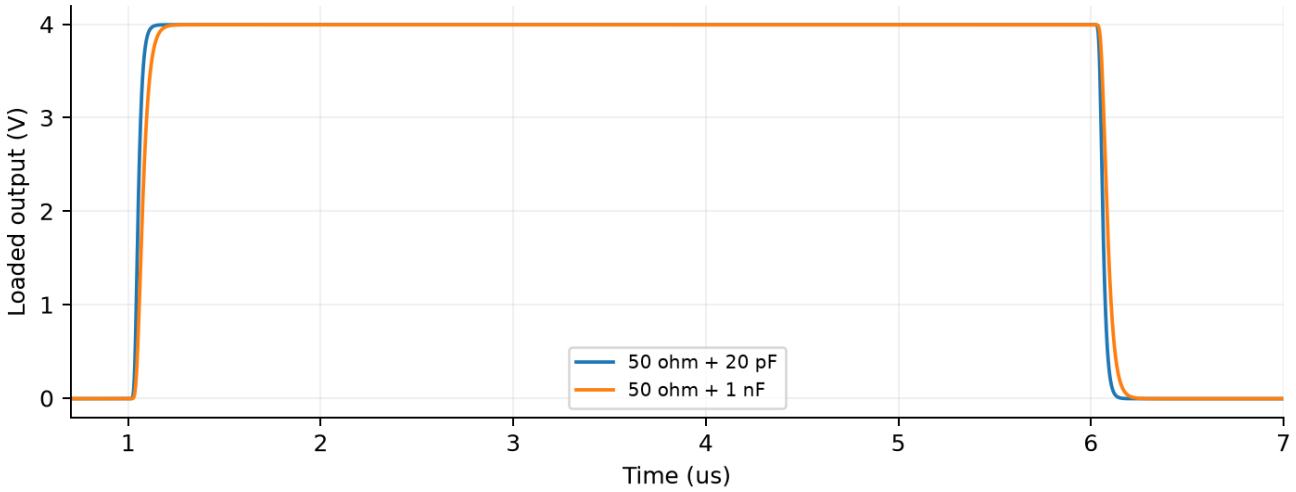
The DAC is represented by a datasheet-derived current-output network plus an assumed 10 MHz first-order pole. TI's available DAC7811 model is encrypted and was not used in LTspice. Switches use static resistance/capacitance models. Thus these results do not guarantee 1% accuracy, cancellation depth, crosstalk, switching glitches, noise, temperature drift or stability on the finished PCB.

An intentionally slower 5 MHz DAC-pole stress gives 2.15% droop at 1 MHz and misses the target. The datasheet bandwidth is typical, not a guaranteed lower limit. Qualify actual assembled boards over frequency and temperature; keep the original design available until that comparison is complete.

Transfer response and settling



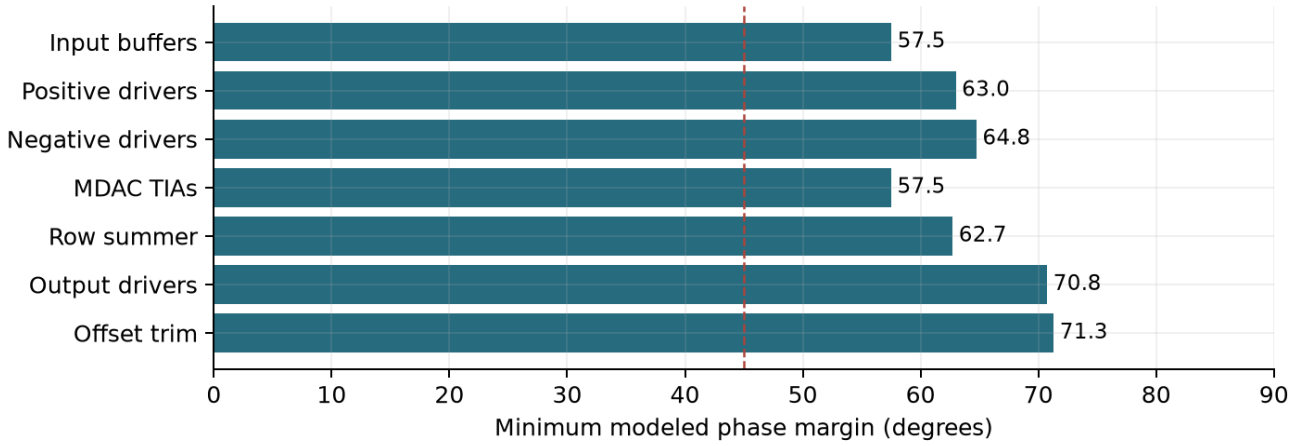
Each curve is normalized to its own DC gain, as after gain calibration. The nominal full-scale coefficient uses q=0.93552 to allow calibration headroom. The resistor/capacitance corner uses 12 kohm DAC resistance, 20 pF total summing-node capacitance, 1.5 pF total feedback capacitance, and 245 ohm switch resistance.



All four inputs step from 0 to +1 V together and return to zero. The nominal output settles to 3.9945 V (uncorrected modeled DC offset included). The recorded 1% rise settling times are 118 ns with 20 pF and 181 ns with 1 nF. Both transient runs reach 12 us; no sustained ringing is present in these models.

DC sweeps cover both input polarities and both coefficient signs. The fitted slope is within 0.001% of unity in the ideal calibrated model; the largest fitted residual is below 1.6 uV. This is a model result, not a tolerance or measurement claim. The typical modeled untrimmed output offset is about -5.5 mV.

Feedback stability and design limits



Loop gain is measured with an AC series injection while retaining the DC operating point. Phase margins are evaluated at downward unity-gain crossings. The input loops include all four reference loads. Twelve TIA cases cover 8/10/12 kohm feedback resistance, 10/20 pF input capacitance, and 1.1/1.5 pF total feedback capacitance. Output-driver cases include 20 pF and 1 nF loads behind the 49.9 ohm source resistor.

Use 1.0 pF C0G +/-0.1 pF TIA compensation capacitors and short feedback paths. Do not substitute a slower amplifier or change a current-feedback driver's feedback resistor without repeating the analysis. Plane exclusions around sensitive summing pads reduce capacitance; final parasitics require hardware checks.

Accuracy and calibration

Twelve-bit coefficient magnitude gives approximately 0.026% coefficient steps after the analog gain reserve. Calibrate the positive and negative gain of every coefficient separately with the actual 50 ohm measurement load. Trim each output at zero input; the nominal trim step is about 55 uV at the loaded connector. Small-coefficient relative error is not a guaranteed specification.

Bench qualification

Start muted with current-limited analog rails, no input signal, and installed insulated regulator heatsinks. Check all rails and offsets, then calibrate DC gain and offset. Sweep each matrix path through 1 MHz, check all-positive/all-negative sums at full amplitude, inspect output and internal amplifier nodes for oscillation, and run a full-load thermal soak inside the enclosure. Check power sequencing and cable loading. USB and analog grounds are common.

Reproduction and primary references

LTspice 24.1.3; vendor amplifier macromodels plus documented behavioral DAC/switch elements. Transients use Gear integration. Sources, checksums, benches, results and the report generator are stored alongside this PDF. Vendor models are downloaded separately subject to their licenses.

Manufacturer datasheets: DAC7811 | OPA2810 | TMUX6136 | OPA4197 family | MCP4728